Light

GCSE Computer Science - Programming Fundamentals (Set 2)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the difference between the Character and String data types?

- ○ **A.** Character stores a single character while String can store zero or more characters
- ○ **B.** Character stores numbers while String stores letters
- ○ **C.** Character is for uppercase letters while String is for lowercase letters
- ○ **D.** There is no difference between Character and String

Q2 (1 mark)

What does the ^ operator do in programming?

- ○ **A.** Performs multiplication
- ○ **B.** Performs division
- ○ **C.** Raises a number to a power (exponentiation)
- ○ **D.** Calculates the remainder after division

Q3 (1 mark)

Which statement is TRUE when x = 5?

- ○ **A.** x != 5
- ○ **B.** x != 10
- ○ **C.** x != 5 AND x != 10
- ○ **D.** x == 10

Q4 (1 mark)

What is a global variable?

- ○ **A.** A variable that can only be accessed within one subroutine
- ○ **B.** A variable that is automatically deleted after use
- ○ **C.** A variable that can be accessed from any part of the program
- ○ **D.** A variable that stores website addresses

Q5 (1 mark)

What does the input() function do?

- ○ **A.** Receives data from the user and stores it in a variable
- ○ **B.** Displays a message on the screen
- ○ **C.** Performs a calculation on numbers
- ○ **D.** Writes data to a file

Q6 (1 mark)

What does print("Hello World") do?

- ○ **A.** Stores "Hello World" in a variable
- ○ **B.** Reads "Hello World" from a file
- ○ **C.** Displays "Hello World" on the screen
- ○ **D.** Sends "Hello World" to a printer device

Q7 (1 mark)

What is the value of result?

```
word = "hello"
result = word.upper
```

- ○ **A.** "HELLO"
- ○ **B.** "hello"

- ○ **C.** "Hello"
- ○ **D.** 5

Q8 (1 mark)

What type of loop uses a step value to change the counter by a specified amount each iteration?

- ○ **A.** While loop
- ○ **B.** Do-until loop
- ○ **C.** For loop with step
- ○ **D.** Infinite loop

Q9 (1 mark)

How many times will the loop body execute at minimum in a do-until loop?

- ○ **A.** One time
- ○ **B.** Zero times
- ○ **C.** Two times
- ○ **D.** It depends on the condition

Q10 (1 mark)

What does the <= operator check?

- ○ **A.** If the left value is strictly less than the right value
- ○ **B.** If the left value is less than or equal to the right value
- ○ **C.** If the left value is greater than the right value
- ○ **D.** If the values are not equal

Q11 (1 mark)

What is the value of result?

```
message = "Testing"
result = message.length
```

- ○ **A.** 6
- ○ **B.** 7
- ○ **C.** 8
- ○ **D.** "Testing"

Q12 (1 mark)

What is the result of NOT True?

- ○ **A.** False
- ○ **B.** True
- ○ **C.** Error

- ○ **D.** Null

Q13 (2 marks)

A program needs to store a customer's postcode, their account balance in pounds and pence, and whether their account is active. Which combination of data types is most appropriate?

- ○ **A.** Integer, Real, Boolean
- ○ **B.** Real, String, Boolean
- ○ **C.** String, Integer, String
- ○ **D.** String, Real, Boolean

Q14 (2 marks)

What is the value of result?

```
result = 3^4
```

- ○ **A.** 12
- ○ **B.** 81
- ○ **C.** 7
- ○ **D.** 64

Q15 (2 marks)

How many times will "Count" be printed?

```
for i = 2 to 10 step 2
    print("Count")
next i
```

- A. 4
- B. 8
- C. 10
- D. 5

Q16 (2 marks)

What is output by this code when grade = "B"?

```
switch grade:
    case "A":
        print("Excellent")
    case "B":
        print("Good")
    case "C":
        print("Satisfactory")
    default:
        print("Needs work")
endswitch
```

- A. Excellent
- B. Good
- C. Satisfactory
- D. Needs work

Q17 (2 marks)

What is the value of result?

```
text = "COMPUTING"
result = text.substring(3, 4)
```

- A. "COMP"
- B. "MPUT"
- C. "PUT"
- D. "PUTI"

Q18 (2 marks)

What is the value of result?

```
word = "PROGRAMMING"
result = word.right(6)
```

- A. "PROGRA"
- B. "MMING"
- C. "AMMING"
- D. "GRAMMI"

Q19 (2 marks)

What is output by this code?

```
procedure greet(name)
    print("Hello " + name)
endprocedure

greet("Alice")
```

- A. Hello name
- B. Error
- C. greet Alice
- D. Hello Alice

Q20 (2 marks)

What is output by this code?

```
global counter = 10

procedure addFive()
    counter = counter + 5
endprocedure

addFive()
print(counter)
```

- A. 10
- B. 15
- C. 5
- D. Error

Q21 (2 marks)

What is the value of total?

```
num1 = "25"
num2 = "30"
total = int(num1) + int(num2)
```

- A. "2530"
- B. Error
- C. "55"
- D. 55

Q22 (2 marks)

Which file operation is used to add data to a file?

- A. readLine()
- B. writeLine()
- C. open()
- D. close()

Q23 (2 marks)

What value is stored at numbers[3] after this code runs?

```
array numbers[5]
numbers[0] = 10
numbers[1] = 20
```

```
numbers[2] = 30
numbers[3] = 40
numbers[4] = 50
```

- A. 30
- B. 4
- C. 50
- D. 40

Q24 (3 marks)

What is the final value of sum after this code executes?

```
sum = 0
for i = 10 to 2 step -2
    sum = sum + i
next i
```

- A. 30
- B. 20
- C. 40
- D. 55

Q25 (3 marks)

What is output by this code?

```
array grid[3,4]
grid[0,0] = "A"
grid[0,1] = "B"
```

```
grid[1,0] = "C"
grid[1,1] = "D"
grid[2,0] = "E"
grid[2,1] = "F"
print(grid[1,1])
```

- ○ **A.** "A"
- ○ **B.** "C"
- ○ **C.** "D"
- ○ **D.** "F"

Q26 (3 marks)

A programmer needs a subroutine that calculates the area of a rectangle and uses the result in a mathematical expression. Should they use a function or a procedure, and why?

- ○ **A.** A procedure, because procedures are faster than functions
- ○ **B.** A function, because functions can return values that can be used in expressions
- ○ **C.** A procedure, because procedures can take parameters
- ○ **D.** Either could work equally well for this task

Q27 (3 marks)

What is the value of result?

```
a = 5
b = 10
```

```
c = 15
result = NOT(a > b) OR (c > b)
```

- A. True
- B. False
- C. 5
- D. 15

Q28 (3 marks)

How many times will "X" be printed?

```
for i = 1 to 3
    for j = 1 to 4
        print("X")
    next j
next i
```

- A. 7
- B. 3
- C. 4
- D. 12

Q29 (3 marks)

Which SQL statement correctly retrieves all products that are either in the 'Electronics' category OR have a price less than 50?

- A. SELECT * FROM Products WHERE category = "Electronics" AND

price < 50

○ **B.** GET * FROM Products IF category = "Electronics" OR price < 50

○ **C.** SELECT * FROM Products WHERE category = "Electronics" OR price < 50

○ **D.** SELECT category, price FROM Products WHERE Electronics OR 50

Q30 (3 marks)

What does this code do?

```
myFile = open("data.txt")
while NOT myFile.endOfFile()
    line = myFile.readLine()
    print(line)
endwhile
myFile.close()
```

○ **A.** Writes new data to the file

○ **B.** Reads and displays every line from the file until the end is reached

○ **C.** Deletes the contents of the file

○ **D.** Creates a new file called data.txt

Total: **55** marks • Time: **30 min**

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